

The Alaoglu–Erdős Integer-Exponent Conjecture

Repository Audit, Exact Conditional Structure,
and New Partial Results

A detailed analysis of six commits in

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Bottom line. No unconditional proof of the full two-base conjecture is obtained. The investigation does, however, produce a substantially stronger conditional structure theorem: if one counterexample exists, there is a unique least nonintegral solution β , and the complete solution set is exactly

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\}.$$

Equivalently, for $M = 2^\beta$ and $A = 3^\beta$, every candidate pair is uniquely $(2^n M^k, 3^n A^k)$. This gives exact counts in every dyadic block and an immediate Weyl-equidistribution theorem. The report also proves a sharper family of finite-difference obstructions, audits the precise Lean material present at the requested commit, and gives an independent interval-arithmetic exclusion of all nonintegral candidates with $2^x < 2^{20}$. That last computation is explicitly *not* claimed to be a Lean kernel proof.

August 20, 2026

Abstract

The Alaoglu–Erdős conjecture asks whether a real x satisfying $2^x, 3^x \in \mathbb{Z}$ must be integral. It is a particularly concrete unresolved instance of the four exponentials problem. This report examines commits `f492c18`, `5ed4e61`, `a37d97f`, `4a9b03d`, `18a85e6`, and `423ac24` in the `ProveIt` repository, together with `two-base-partial-results.tex`.

The source audit confirms a substantial committed Lean corpus: the three-base theorem for 2, 3, 5; irrationality and transcendence restrictions; verified exclusion below $2^x = 256$; a rank-at-most-two theorem for prime-exponent vectors; logarithmic-square counting; zero-density and local progression obstructions; affine descent; shrinking-target bounds; and classification of additional integral bases as 3-smooth. It also identifies a reproducibility discrepancy: at commit `423ac24`, the survey cites `FiniteCheck4096`, `OddCore`, and `FractionalPartDensity`, but those files are not present in that commit tree and are not on its import surface. Results depending on them are therefore treated here as article-level claims, not as audited committed Lean.

Starting from the committed cross-product identity for odd prime valuations, we prove a new exact structure theorem. Conditional on a counterexample, all natural candidates have one primitive odd core w , and a rational-power subgroup argument removes the residual semigroup ambiguity. There are canonical integers $a \geq 0$ and $d \geq 1$ such that

$$\beta = a + d \log_2 w, \quad 2^\beta = M = 2^a w^d, \quad 3^\beta = A \in \mathbb{N},$$

and every solution has the unique form $n + k\beta$. Consequently, the candidate monoid is freely generated by 2 and M , the number of nonintegral solutions in $[N, N+1)$ is exactly $\lfloor (N+1)/\beta \rfloor$, and their fractional parts in a high unit interval are an initial segment of the irrational rotation $k\beta \pmod{1}$, hence become equidistributed.

We also prove that a progression $n, n+h, \dots, n+rh$ of candidate bases is impossible whenever

$$|\vartheta(\vartheta-1) \cdots (\vartheta-r+1)| h^r < n^{r-\vartheta}, \quad \vartheta = \log_2 3,$$

strictly improving the committed $h^5 \leq n$ three-term criterion and extending it to arbitrary length. Finally, a reproducible 40-decimal interval computation checks every non-power of two $m < 2^{20}$ and certifies $m^\vartheta \notin \mathbb{Z}$. This raises the independently checked finite barrier from the committed $x \geq 8$ to $x \geq 20$, subject to the stated software trust boundary. The remaining obstruction is isolated as a radical four-exponentials statement: for no odd $u > 1$ should u^ϑ lie in $\mathbb{Z}[1/3]$.

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1 Executive conclusions

1.1 What was and was not proved

The full conjecture remains unresolved in this investigation. That conclusion is not a premature appeal to its historical status: several routes were pushed until their exact failure point became visible. In particular, the repository’s determinant method was examined for a two-by-two replacement; the multiplicative-rank statements were converted into an exact conditional classification; the local convexity argument was sharpened to all higher finite differences; and the finite search was independently extended. Each route leaves the same transcendence-theoretic core rather than an elementary loose end.

The strongest new mathematical conclusion is the following conditional theorem.

Theorem 1.1 (Exact primitive-generator theorem; informal executive form). *Assume there exists a nonintegral real x with $2^x, 3^x \in \mathbb{Z}$. Then there is a unique least nonintegral such number $\beta > 0$. If $M = 2^\beta$ and $A = 3^\beta$, then*

$$\mathcal{S} := \{x \in \mathbb{R} : 2^x, 3^x \in \mathbb{Z}\} = \{n + k\beta : n, k \in \mathbb{N}_0\},$$

and the representation is unique. Equivalently,

$$\{(2^x, 3^x) : x \in \mathcal{S}\} = \{(2^n M^k, 3^n A^k) : n, k \in \mathbb{N}_0\}.$$

Status: [new proof in this report]

A complete proof appears in [section 7](#). It uses the committed Lean theorem `odd_factorization_cross_mul` as an input, but the deduction from that theorem to the exact monoid classification has not been checked by Lean in the audited repository.

This removes a misleading degree of freedom from the survey’s proposed “exponent semigroup” direction. Conditional on failure, the set of odd-core exponents is not an arbitrary additive submonoid with possible Frobenius gaps. After primitive normalization it is exactly $d\mathbb{N}_0$, and the required power of two is governed by one linear threshold.

1.2 Four evidence levels

To avoid merging unlike kinds of evidence, every important result is assigned one of four labels.

[committed Lean]

A proof source is present in the requested commit tree and lies on the audited `ExponentialIdentities.lean` import surface. This report inspected the source but did not independently rebuild the repository with Lean.

[article-only at the audited commit]

The survey states the result or lists a module, but the corresponding file is absent from the requested commit tree, or the survey itself marks the result as pending.

[new proof in this report]

A conventional mathematical proof is supplied in this report. It is intended to be suitable for formalization, but no claim of kernel checking is made.

[interval computation]

A finite statement was checked by outward-rounded interval arithmetic. The script, parameters, worst margins, and hashes are recorded. The trusted base includes Python and `mpmath`; this is not interchangeable with a Lean certificate.

1.3 Main deliverables

The report provides the following concrete advances.

1. A commit-by-commit reconstruction of how the formal development moved from the three-base theorem to the two-base partial-results corpus and then to the survey.
2. A precise audit of which cited modules and theorems are actually present at `423ac24`.
3. A full proof of the primitive odd-core lemma from the committed cross-valuation identity.
4. The exact primitive-generator theorem, exact candidate monoid, and exact output-pair monoid.
5. Exact dyadic-block and cumulative counting formulas, improving the qualitative and coarse logarithmic-square bounds conditionally on failure.
6. Equidistribution of fractional parts in each high unit interval.
7. A one-number localization reformulation and a canonical radical-algebraic reformulation of any counterexample.
8. A sharper arbitrary-order finite-difference obstruction for arithmetic progressions of candidate bases.
9. An independent interval certificate through $m < 2^{20}$, yielding $x > 20$ for any counterexample under that software trust model.
10. A prioritized Lean formalization and research program focused on statements that can plausibly move the boundary rather than merely restating the conjecture.

2 The problem and its equivalent forms

2.1 Original statement

Conjecture 2.1 (Alaoglu–Erdős, two-base integer form). *For every $x \in \mathbb{R}$,*

$$2^x \in \mathbb{Z} \quad \text{and} \quad 3^x \in \mathbb{Z} \quad \implies \quad x \in \mathbb{Z}.$$

Since 2^x and 3^x are positive, the hypotheses actually put them in $\mathbb{N}$. Moreover, $2^x \in \mathbb{N}$ forces $x \geq 0$: a negative exponent gives $0 < 2^x < 1$. Thus all solutions live in $[0, \infty)$.

Alaoglu and Erdős formulated the broader expectation that rational powers of two distinct primes should not simultaneously be rational at a nonintegral exponent, in the course of their study of highly composite and related numbers [1]. The modern two-base integer form is both elementary to state and a direct special case of the four exponentials conjecture. As of the dated literature check for this report, the four exponentials conjecture remains open [8, 9].

2.2 Candidate-base normalization

Put

$$\vartheta := \frac{\log 3}{\log 2} = \log_2 3 = 1.584962500721156\dots$$

and define

$$\mathcal{C} := \{m \in \mathbb{N}_{>0} : m^\vartheta \in \mathbb{N}\}.$$

Lemma 2.2 (Exponent–candidate bijection). *The maps*

$$x \mapsto 2^x, \quad m \mapsto \log_2 m$$

are mutually inverse bijections between

$$\mathcal{S} := \{x \in \mathbb{R} : 2^x, 3^x \in \mathbb{Z}\} \quad \text{and} \quad \mathcal{C}.$$

Under the bijection, $3^x = m^\vartheta$.

Proof. If $x \in \mathcal{S}$, set $m = 2^x \in \mathbb{N}$. Then

$$m^\vartheta = \exp\left(\log m \frac{\log 3}{\log 2}\right) = \exp(x \log 3) = 3^x \in \mathbb{N}.$$

Conversely, if $m \in \mathcal{C}$ and $x = \log_2 m$, then $2^x = m$ and the same identity gives $3^x = m^\vartheta \in \mathbb{N}$. $\square$

Integral solutions correspond exactly to powers of two. Therefore the conjecture is

$$\mathcal{C} = \{2^n : n \in \mathbb{N}_0\}.$$

The candidate formulation is particularly well suited to unique factorization, finite certification, finite differences of t^ϑ , and multiplicative-rank arguments.

2.3 The four-exponentials reduction

Suppose x is nonintegral and $2^x, 3^x$ are algebraic. The rational-exponent argument shows that x cannot be rational; hence $1, x$ are linearly independent over $\mathbb{Q}$. The numbers $\log 2, \log 3$ are also $\mathbb{Q}$ -linearly independent, since $2^a 3^b = 1$ with integers a, b forces $a = b = 0$. The four exponentials associated to these two pairs are

$$e^{1 \cdot \log 2} = 2, \quad e^{1 \cdot \log 3} = 3, \quad e^{x \cdot \log 2} = 2^x, \quad e^{x \cdot \log 3} = 3^x.$$

All four would be algebraic, contrary to the four exponentials conjecture. This is exactly why the two-base statement lies at the first unresolved 2×2 transcendence threshold, while three multiplicatively independent bases can be handled by the six exponentials theorem.

The reduction is not merely historical context. It is a diagnostic: any proposed elementary argument must exploit special integrality, positivity, or arithmetic structure beyond the bare algebraicity of those four exponentials, or else it would prove a substantial case of the four exponentials conjecture by another name.

2.4 Rational and algebraic exponents

A rational exponent is easy.

Lemma 2.3 (Rational exponent). *If $x \in \mathbb{Q}$ and $2^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$.*

Proof. Write $x = a/b$ in lowest terms with $b > 0$. If $2^{a/b} = m \in \mathbb{N}$, then $2^a = m^b$. Comparing the exponent of the prime 2 gives $a = b v_2(m)$, so $b \mid a$ and $b = 1$. $\square$

Gelfond–Schneider gives the stronger dichotomy used in the repository: if 2^x is algebraic and x is algebraic irrational, then 2^x is transcendental, a contradiction. Hence a nonintegral solution must be transcendental. This does not resolve the conjecture because the unknown exponent is allowed to be transcendental; indeed, the repository also formalizes that $\vartheta = \log_2 3$ is transcendental.

3 Commit-by-commit reconstruction

The six requested commits form three natural pairs: a feature commit and its merge for the three-base theorem; a feature commit and its merge for the two-base partial corpus; and a survey commit describing that corpus. The table records the role of each revision rather than merely its message.

Commit	Recorded purpose	Mathematical and repository role
f492c18	Formalize integer real exponent theorem	Introduces the determinant/grid/kernel development and proves that integrality of $2^x, 3^x, 5^x$ forces $x \in \mathbb{Z}$. The irrational case is excluded by an interpolation determinant; the rational case is discharged by prime factorization.
5ed4e61	Merge the three-base branch	Integrates the preceding development into the main history. Its second parent is f492c18; mathematically it carries the same three-base result.
a37d97f	Formalize partial results for two-base integer exponents	Adds the basic two-base module: the conjecture as a proposition, closure under natural multiples, irrationality of a nonintegral candidate, endpoint gaps for fractional parts, initial numerical bounds for ϑ , a small finite exclusion, and a local second-difference obstruction.

Commit	Recorded purpose	Mathematical and repository role
4a9b03d	Formalize strong partial results	Adds the major corpus: finite checks through 256, arithmetic descent, multiplicative rank, candidate structure, density results, shrinking targets, the 3-smooth theorem, a four-exponentials reduction, transcendence consequences, and a substantial Gelfond–Schneider port.
18a85e6	Merge the two-base branch	Merge revision carrying the same tree-level mathematical content as 4a9b03d.
423ac24	Survey partial results	Adds <code>two-base-partial-results.tex</code> and its compiled PDF. The survey synthesizes the formal corpus, but also describes three modules that are not present in this commit tree; this requires a careful evidence split discussed below.

3.1 The determinant architecture

The three-base proof in `IntegerExponent.lean` is the conceptual foundation for several later results. In outline, for an irrational x it constructs a large square matrix

$$\mathcal{A}_{ij} = \exp(a_i b_j),$$

where the row arguments encode monomials in $2, 3, 5$ and the column arguments encode affine forms in $1, x$. Irrationality gives injectivity of the column parameters; multiplicative independence gives injectivity of the row parameters. Total positivity/nonvanishing of the exponential determinant gives $\det \mathcal{A} \neq 0$. The integrality hypotheses make every entry integral, so the nonzero determinant has absolute value at least one. An interpolation remainder estimate, with explicit choices of the matrix size and truncation order, gives the contradictory upper bound $|\det \mathcal{A}| < 1$.

This is a formal, specialized realization of the six-exponentials mechanism. The later `MultiplicativeRank.lean` abstracts the row bases: three natural candidates with linearly independent prime-exponent vectors would give three multiplicatively independent bases and therefore force ϑ to be rational, contradicting its irrationality. This turns transcendence into a rank bound in the free abelian group on the primes.

3.2 Why the same determinant does not simply shrink to two bases

The dimension count is critical. With three independent row directions and two independent column directions, one can choose enough interpolation conditions to make the analytic upper bound beat the arithmetic lower bound. In the two-by-two case the determinant estimate lands at the classical boundary where the known proof no longer yields decay below one. Changing constants, enlarging the finite grid, or optimizing the same asymptotic balance does not remove this structural deficit. A successful two-base determinant proof would amount to a new four-exponentials argument or would have to use the fact that the four algebraic values are actually positive integers in a way absent from the general theorem.

4 Audit of the formal corpus at 423ac24

4.1 Audit convention

This was a source-tree audit of the exact requested revision, not an independent kernel replay. A module is counted as present when its file can be fetched at 423ac24; a theorem is counted as committed when its proof source appears there; and a module is counted as part of the advertised import surface when it is imported by `Analysis/ExponentialIdentities/Lean/ExponentialIdentities.lean`.

The imported two-base modules at that revision are:

```
TwoBaseIntegerExponent, FiniteCheck, FiniteCheck256, ArithmeticRigidity,
FourExponentialsReduction, ThreeSmooth, ZeroDensity, MultiplicativeRank,
CandidateStructure, NonintegerDensity, ShrinkingTarget, Transcendence.
```

The facade also imports `IntegerExponent` and `TinyExponentTower`.

4.2 Results visibly present in committed Lean

The following list is selective but covers the main logical axes.

Module / theorem family	Audited content
<code>IntegerExponent</code>	$2^x, 3^x, 5^x \in \mathbb{Z} \Rightarrow x \in \mathbb{Z}$; rationality under the three hypotheses; the rational exponent lemma; determinant lower and upper bounds.
<code>TwoBaseIntegerExponent</code>	Definition of the conjecture; closure under natural multiples; nonintegral candidate implies irrational exponent; explicit endpoint gaps; bounds $1 < \vartheta < 8/5$; local second-difference arguments.
<code>FiniteCheck, FiniteCheck256</code>	Kernel-oriented finite certificates through $2^x < 100$ and $2^x < 256$; in exponent coordinates, any nonintegral solution satisfies $x \geq 8$.
<code>ArithmeticRigidity</code>	Common-perfect-power and affine descent; if outputs have forms $2^r u^k$ and $3^r v^k$, then $(x - r)/k$ is another solution; combining with the finite region gives $r + 8k \leq x$.
<code>MultiplicativeRank</code>	Any three candidate prime-exponent vectors are linearly dependent over $\mathbb{Q}$; every finite candidate set has rank at most two; at most $(\log_2 N)^2$ candidates lie below N .
<code>CandidateStructure</code>	For candidates a, b , an odd prime $p \mid a$, and odd prime coordinate q , the cross-product identity $v_p(a)v_q(b) = v_p(b)v_q(a)$; fixed-power relations; conditional quadratic lower boxes if a non-power-of-two candidate exists.
<code>ZeroDensity,</code> <code>NonintegerDensity</code>	A local three-term-progression obstruction and density-zero conclusions for candidate and noninteger-candidate sets.

Module / theorem family	Audited content
ShrinkingTarget	Explicit lower bounds on the distance of kx from integers for every positive k , including the dyadic shrinking radius of exponential order 2^{-kx} .
ThreeSmooth	A third multiplicatively independent natural base with integral x -th power forces $x \in \mathbb{Z}$; under a hypothetical counterexample, the positive natural bases a with $a^x \in \mathbb{Z}$ are exactly the 3-smooth numbers $2^u 3^v$.
Transcendence	If $2^x \in \mathbb{Z}$, then x is integral or transcendental; in particular every counterexample is transcendental; $\vartheta = \log_2 3$ is transcendental.
FourExponentialsReduction	A formal implication from the appropriate four-exponentials statement to the two-base conjecture.

4.3 Three survey modules absent from the commit tree

The survey’s summary table and formalization inventory name

`FiniteCheck4096.lean`, `OddCore.lean`, `FractionalPartDensity.lean`.

Direct lookup at 423ac24 returns no such files, and none appears in the facade import list. The consequences are different in the three cases.

- The text itself says the 4096 certificate is pending final Lean acceptance. The claimed exponent barrier $x \geq 12$ should therefore be read as a proposed or externally generated certificate at this revision, not a committed theorem.
- The common-odd-core theorem is used throughout the survey and listed as formalized, but the named module is absent. Fortunately, its main mathematical statement can be derived from the committed `odd_factorization_cross_mul`; [section 6](#) supplies a proof and then strengthens it.
- The fractional-part density, gap equivalences, and infinitude statements are likewise attributed to an absent module. The exact generator theorem below yields stronger density and equidistribution conclusions as ordinary proofs.

This is a reproducibility observation about one exact revision, not a claim that the omitted files never existed on another branch or machine. The appropriate correction is simply to separate the evidence levels instead of allowing the prose inventory to upgrade absent source into audited formal proof.

5 What the committed results already force

Suppose for the rest of this section that x is a counterexample and write

$$m = 2^x \in \mathbb{N}, \quad n = 3^x \in \mathbb{N}.$$

The existing corpus imposes the following simultaneous restrictions.

5.1 Transcendence and size

The exponent x is irrational, indeed transcendental. The committed finite certificate gives $x \geq 8$; because x is nonintegral, actually $x > 8$. Equivalently, $m \geq 256$ and m is not a power of two.

5.2 Multiplicative rank

Every natural candidate a gives a prime-exponent vector

$$\mathbf{v}(a) = (v_p(a))_{p \text{ prime}} \in \mathbb{Q}^{(\mathbb{P})}.$$

Any three such vectors are linearly dependent, so the rational span of every finite candidate set has dimension at most two. Since 2 is a candidate, all odd parts must lie on at most one additional rational direction. This is the doorway to the exact odd-core theorem below.

5.3 Local gaps and shrinking targets

For every $k \geq 1$, kx is again a solution. Its fractional part cannot be too close to either endpoint of its unit interval. In a simplified form, if $y = kx$ then

$$\|y\|_{\mathbb{Z}} \gtrsim 2^{-y}.$$

The repository gives exact logarithmic radii rather than only this asymptotic notation. Such a bound is compatible with ordinary Diophantine approximation: Dirichlet guarantees infinitely many k with $\|kx\|_{\mathbb{Z}} < 1/k$, and 2^{-kx} is incomparably smaller. Thus density of fractional parts does not by itself contradict the shrinking target.

5.4 Affine descent

If

$$m = 2^r u^k, \quad n = 3^r v^k$$

with $r \geq 0$ and $k \geq 1$, then $(x - r)/k$ is another solution. Since committed finite checking excludes nonintegral solutions below 8, one obtains

$$r + 8k \leq x.$$

For a general counterexample this is quantitative. For the least counterexample developed below, the same descent becomes an exact primitivity statement: no nontrivial such decomposition can exist at all.

5.5 Additional bases

If $a \in \mathbb{N}_{>0}$ and $a^x \in \mathbb{Z}$, then $a = 2^u 3^v$. Conversely every such 3-smooth base has integral x -th power. This is an exact classification of bases for a *fixed* hypothetical counterexample. It should not be confused with classification of the candidate values 2^x as x ranges over all solutions; the latter is the subject of [section 7](#).

6 A primitive common odd core

This section proves, at paper level, the principal statement attributed in the survey to the missing `OddCore.lean`. The proof starts from the committed cross-valuation theorem and uses only finite-support unique factorization and Bézout.

6.1 Odd valuation profiles lie on one ray

Lemma 6.1 (Cross-valuation input). *Let $a, b \in \mathcal{C}$. If p is an odd prime with $v_p(a) > 0$, then for every odd prime q ,*

$$v_p(a)v_q(b) = v_p(b)v_q(a).$$

Status: [\[committed Lean\]](#)

This is the mathematical content of `odd_factorization_cross_mul` in `CandidateStructure.lean`.

Theorem 6.2 (Primitive common odd core). *Assume $\mathcal{C}$ contains a non-power of two. Then there is a unique odd integer $w > 1$ that is power-free in the sense*

$$\gcd\{v_p(w) : p \mid w\} = 1,$$

and every candidate $b \in \mathcal{C}$ has a unique representation

$$b = 2^i w^j, \quad i, j \in \mathbb{N}_0.$$

Proof. Choose a non-power-of-two candidate a . Its odd valuation vector $A = (A_q)_{q \neq 2}$, $A_q = v_q(a)$, is nonzero and finitely supported. Let

$$g = \gcd\{A_q : A_q > 0\}, \quad c_q = A_q/g, \quad w = \prod_{q \neq 2} q^{c_q}.$$

Then w is odd, greater than one, and the positive c_q have gcd one.

Let $b \in \mathcal{C}$ and set $B_q = v_q(b)$. Fix an odd prime p with $A_p > 0$. By [theorem 6.1](#),

$$A_p B_q = B_p A_q \quad (q \neq 2).$$

If $A_q = 0$, this forces $B_q = 0$. On the support of A it says

$$B_q = \frac{B_p}{A_p} A_q = \left(\frac{g B_p}{A_p} \right) c_q.$$

Put $t = g B_p / A_p \in \mathbb{Q}_{\geq 0}$. Every $t c_q$ is an integer. Since the c_q have gcd one, there exist integers r_q , only finitely many nonzero, with $\sum r_q c_q = 1$. Therefore

$$t = \sum_q r_q (t c_q) \in \mathbb{Z}.$$

Write $j = t \in \mathbb{N}_0$. The odd part of b is exactly w^j , so $b = 2^i w^j$ with $i = v_2(b)$.

Uniqueness for fixed w follows from odd valuations and then from the 2-adic valuation. If w' is another primitive common core, the odd valuation vectors of w and w' generate the same rational ray and are both primitive integer vectors; hence they agree. $\square$

Status: [new proof in this report]

The theorem is stronger in normalization than a bare common-core statement: taking the gcd of the odd valuation profile makes w canonical and prevents hidden perfect powers from being absorbed into its exponent.

6.2 The logarithmic lattice and its residual ambiguity

Set

$$\alpha := \log_2 w, \quad E := 3^\alpha = w^\vartheta.$$

Then every candidate exponent has the form

$$x = \log_2(2^i w^j) = i + j\alpha.$$

The number α is irrational: if it were rational, the rational-exponent lemma applied to $2^\alpha = w \in \mathbb{Z}$ would make α integral, forcing the odd $w > 1$ to be a power of two. In fact α is transcendental by the committed Gelfond–Schneider corollary.

At this stage the common-core statement alone does *not* tell us which pairs (i, j) occur. The candidate condition is

$$3^{i+j\alpha} = 3^i E^j \in \mathbb{N}.$$

The survey leaves these pairs in an additive semigroup $B \subseteq \mathbb{N}_0^2$. The next section shows that B has a completely rigid form.

7 Exact conditional structure of all solutions

7.1 The rational-power index

Because a non-power-of-two candidate exists, there is some $j > 0$ such that E^j is rational: indeed, $3^j E^j \in \mathbb{N}$. Let

$$d := \min\{j \in \mathbb{N}_{>0} : E^j \in \mathbb{Q}\}.$$

The following elementary fact is the key step.

Lemma 7.1 (All rational powers are multiples of the least one). *For every $j \in \mathbb{Z}$,*

$$E^j \in \mathbb{Q} \iff d \mid j.$$

Proof. The set

$$J := \{j \in \mathbb{Z} : E^j \in \mathbb{Q}\}$$

is an additive subgroup of $\mathbb{Z}$: $0 \in J$, products add exponents, and reciprocals negate them. Every nonzero subgroup of $\mathbb{Z}$ is $d\mathbb{Z}$ for its least positive element d . $\square$

Write the positive rational E^d in lowest terms. Its denominator has no prime other than 3. To see this, take any original candidate with odd-core exponent $j = dk > 0$ and power-of-two exponent i . Since

$$3^i (E^d)^k \in \mathbb{N},$$

any prime in the reduced denominator of E^d would survive in the denominator of its k -th power unless it were 3. Hence there are unique integers $a \geq 0$ and $A \geq 1$ such that

$$E^d = \frac{A}{3^a}, \quad (1)$$

with $3 \nmid A$ when $a > 0$.

7.2 The primitive generator

Define

$$\beta := a + d\alpha = a + d \log_2 w, \quad M := 2^\beta = 2^a w^d, \quad A := 3^\beta. \quad (2)$$

The last notation is consistent with (1), because

$$3^\beta = 3^a(E^d) = A.$$

Thus β is itself a solution. It is irrational because α is irrational.

Theorem 7.2 (Exact primitive-generator theorem). *Assume [theorem 2.1](#) is false. With w, d, a, β, M, A as above:*

(i) *every solution has a unique representation*

$$x = n + k\beta, \quad n, k \in \mathbb{N}_0;$$

(ii) *every candidate has a unique representation*

$$m = 2^n M^k, \quad n, k \in \mathbb{N}_0;$$

(iii) *every output pair has a unique representation*

$$(2^x, 3^x) = (2^n M^k, 3^n A^k), \quad n, k \in \mathbb{N}_0;$$

(iv) β *is the unique least nonintegral solution.*

Proof. Let $x \in \mathcal{S}$. By [theorem 6.2](#),

$$2^x = 2^i w^j$$

for unique $i, j \in \mathbb{N}_0$, and therefore $x = i + j\alpha$. Since

$$3^x = 3^i E^j \in \mathbb{N},$$

we have $E^j \in \mathbb{Q}$, so [theorem 7.1](#) gives $j = dk$ for a unique $k \in \mathbb{N}_0$. Using (1),

$$3^x = 3^i (E^d)^k = 3^{i-ak} A^k.$$

Because $A/3^a$ was reduced, this is an integer exactly when $i \geq ak$. Put $n = i - ak \in \mathbb{N}_0$. Then

$$x = i + dk\alpha = n + k(a + d\alpha) = n + k\beta.$$

Conversely, for any $n, k \in \mathbb{N}_0$,

$$2^{n+k\beta} = 2^n M^k \in \mathbb{N}, \quad 3^{n+k\beta} = 3^n A^k \in \mathbb{N},$$

so $n + k\beta \in \mathcal{S}$.

If $n + k\beta = n' + k'\beta$ and $k \neq k'$, then $\beta = (n' - n)/(k - k') \in \mathbb{Q}$, contradiction. Hence $k = k'$ and $n = n'$. This proves (i), and (ii)–(iii) follow by exponentiation. A nonintegral solution has $k \geq 1$; its least possible value is obtained at $(n, k) = (0, 1)$ and equals β . Uniqueness of the least element follows from uniqueness of the representation. $\square$

Status: [new proof in this report]

This theorem is the principal new result of the report. It sharpens both the committed rank-two theorem and the survey’s article-only odd-core description. Conditional on failure, $\mathcal{S}$ is not merely contained in a rank-two additive group: it is the free additive monoid on 1 and one irrational generator β .

7.3 Monoid isomorphisms

The theorem can be packaged as two exact algebraic isomorphisms:

$$(\mathbb{N}_0^2, +) \xrightarrow{\sim} (\mathcal{S}, +), \quad (n, k) \mapsto n + k\beta,$$

and

$$(\mathbb{N}_0^2, +) \xrightarrow{\sim} (\mathcal{C}, \cdot), \quad (n, k) \mapsto 2^n M^k.$$

Similarly the output-pair monoid $\mathcal{P}$ is freely generated by $(2, 3)$ and (M, A) . Thus every multiplicative relation among candidates is the obvious coordinate relation. This is the exact conditional model against which any future density, progression, or descent argument should be tested.

7.4 Primitive output-pair constraints

The leastness of β upgrades the repository’s inequalities to exact exclusions.

Corollary 7.3 (No matched base depth). *For the primitive pair $(M, A) = (2^\beta, 3^\beta)$,*

$$\min\{v_2(M), v_3(A)\} = 0.$$

Equivalently, either M is odd or $3 \nmid A$ (and possibly both).

Proof. If both valuations were positive, $M/2$ and $A/3$ would be integers and

$$2^{\beta-1} = M/2, \quad 3^{\beta-1} = A/3.$$

Then $\beta - 1$ would be a smaller nonintegral solution. $\square$

In the canonical construction, $v_2(M) = a$, and when $a > 0$ the reduced numerator condition in (1) gives $v_3(A) = 0$ directly.

Corollary 7.4 (No common perfect-power degree). *There is no $k \geq 2$ for which both M and A are perfect k -th powers. Equivalently, the gcd of all positive prime exponents occurring in the factorizations of M and A is one.*

Proof. If $M = U^k$ and $A = V^k$, then $2^{\beta/k} = U$ and $3^{\beta/k} = V$. Since β is irrational, β/k is a nonintegral solution smaller than β . $\square$

Corollary 7.5 (Exact affine primitivity). *Suppose*

$$M = 2^r U^k, \quad A = 3^r V^k.$$

After choosing $r \in \mathbb{N}_0$, $k \in \mathbb{N}_{>0}$, and $U, V \in \mathbb{N}$. Then necessarily $r = 0$ and $k = 1$.

Proof. The affine descent identity gives a solution $(\beta - r)/k$. It is nonintegral because β is irrational. If $r > 0$ or $k > 1$, it is strictly smaller than β , contradicting minimality. $\square$

This corollary is the primitive form of `base_depth_add_eight_mul_degree_le`. For arbitrary solutions the committed theorem gives a lower bound; for the canonical generator, leastness removes every nontrivial affine descent.

8 Equivalent one-number and radical formulations

8.1 Localization at the prime three

The conjecture can be reduced to a statement about one odd integer and one irrational power.

Theorem 8.1 (Localization reformulation). *The Alaoglu–Erdős conjecture is false if and only if there exists an odd integer $u > 1$ such that*

$$u^\vartheta \in \mathbb{Z}[1/3].$$

More explicitly, failure is equivalent to the existence of $u > 1$ odd and $a \in \mathbb{N}_0$ for which $3^a u^\vartheta \in \mathbb{N}$.

Proof. If x is a counterexample, write the integer 2^x uniquely as $2^a u$ with u odd. Since 2^x is not a power of two, $u > 1$. Then

$$u^\vartheta = \frac{(2^x)^\vartheta}{(2^a)^\vartheta} = \frac{3^x}{3^a} \in \mathbb{Z}[1/3].$$

Conversely, if $u^\vartheta = B/3^a$ with $B \in \mathbb{N}$, set $x = a + \log_2 u$. Then

$$2^x = 2^a u \in \mathbb{N}, \quad 3^x = 3^a u^\vartheta = B \in \mathbb{N}.$$

The exponent is not integral, because otherwise the odd $u = 2^{x-a} > 1$ would be a power of two. $\square$

Status: [new proof in this report]

This is a compact target for future work: prove $u^{\log_2 3} \notin \mathbb{Z}[1/3]$ for every odd $u > 1$. It is stronger than merely proving the irrationality of $3^{\log_2 3}$ for the special odd core $u = 3$, but it exposes exactly the extra denominator allowed by shifting the exponent by an integer.

8.2 Canonical radical algebraicity

In the notation of [theorem 7.2](#), $E = 3^{\log_2 w}$ satisfies

$$E^d = \frac{A}{3^a} \in \mathbb{Q}.$$

This is a positive rational number. The minimality of d has an algebraic consequence.

Proposition 8.2 (Exact radical degree). *The binomial*

$$X^d - \frac{A}{3^a}$$

is irreducible over $\mathbb{Q}$, and E is algebraic of exact degree d .

Proof. For a positive rational q , the standard binomial irreducibility criterion says that $X^d - q$ can be reducible only if q is a p -th power in $\mathbb{Q}$ for some prime $p \mid d$; the additional $4 \mid d$ exceptional case concerns a negative multiple of a fourth power and is irrelevant for $q > 0$. If $q = r^p$ and $p \mid d$, the positive real number $E^{d/p}$ is the rational p -th root of q , contradicting the minimality of d . Therefore the binomial is irreducible and is the minimal polynomial of E . $\square$

A counterexample therefore produces a very special algebraic number

$$E = 3^{\log_2 w}.$$

It arises after starting from a transcendental exponent $\log_2 w$. Taking logarithms of $E^d = A/3^a$ yields

$$d \log w \log 3 = (\log A - a \log 3) \log 2. \quad (3)$$

This is a bilinear relation among logarithms of algebraic numbers. Baker’s theorem controls *linear* forms in logarithms with algebraic coefficients; it does not directly rule out (3). The four exponentials conjecture does. The equation is therefore a precise description of the transcendence-theoretic wall, not a disguised elementary factorization problem.

9 Exact counting and equidistribution

The exact generator theorem turns several qualitative statements into equalities.

9.1 One unit interval at a time

Theorem 9.1 (Exact unit-block count). *Assume a counterexample exists and let β be the primitive generator. For every $N \in \mathbb{N}_0$, the nonintegral solutions in $[N, N+1)$ are precisely*

$$x_{N,k} = \lceil N - k\beta \rceil + k\beta, \quad 1 \leq k \leq \left\lfloor \frac{N+1}{\beta} \right\rfloor.$$

Consequently,

$$\#((\mathcal{S} \setminus \mathbb{N}_0) \cap [N, N+1)) = \left\lfloor \frac{N+1}{\beta} \right\rfloor,$$

and

$$\#(\mathcal{C} \cap [2^N, 2^{N+1})) = 1 + \left\lfloor \frac{N+1}{\beta} \right\rfloor.$$

Proof. Fix $k \geq 1$. There is at most one integer n with $N \leq n + k\beta < N + 1$, because the interval has length one. Such an $n \geq 0$ exists exactly when $k\beta < N + 1$, and then it is $n = \lceil N - k\beta \rceil$. Since β is irrational, $(N + 1)/\beta$ is not an integer, so the admissible positive k are exactly those in the stated range. The one integral solution in the block is N . Apply the bijection $x \mapsto 2^x$ for the candidate count. $\square$

The committed bound $\beta > 8$ immediately gives

$$\#((\mathcal{S} \setminus \mathbb{N}_0) \cap [N, N + 1)) \leq \left\lfloor \frac{N}{8} \right\rfloor.$$

Indeed, $k\beta < N + 1$ implies $8k < N + 1$, and the integer $8k$ is therefore at most N . Under the independent finite computation of [section 11](#), $\beta > 20$ and the right side improves to $\lfloor N/20 \rfloor$.

9.2 Cumulative exact count

For integer $T \geq 1$, let

$$C(T) := \#\{m \in \mathcal{C} : m < 2^T\}.$$

Summing [theorem 9.1](#) gives

$$C(T) = T + \sum_{r=1}^T \left\lfloor \frac{r}{\beta} \right\rfloor, \quad (4)$$

and hence

$$C(T) = \frac{T^2}{2\beta} + O(T). \quad (5)$$

The nonintegral candidates alone have count $\sum_{r=1}^T \lfloor r/\beta \rfloor$.

This is conditionally sharper than the committed universal upper bound $C(T) \leq T^2$ arising from $(\log_2 N)^2$, and sharper than the survey’s article-only $\log_2 N \log_3 N$ coefficient. It also matches the committed conditional quadratic lower box in its correct leading order. With $\beta > 8$, the leading coefficient is strictly less than $1/16$; with the independent $\beta > 20$ barrier it is strictly less than $1/40$.

9.3 Equidistribution in high blocks

The fractional part of $x_{N,k}$ is simply $\{k\beta\}$. Thus a high unit interval contains an initial segment of one irrational rotation orbit.

Theorem 9.2 (Blockwise Weyl equidistribution). *Assume a counterexample exists. For every interval $I \subseteq [0, 1]$ whose boundary has Lebesgue measure zero,*

$$\frac{\#\{x \in (\mathcal{S} \setminus \mathbb{N}_0) \cap [N, N + 1) : \{x\} \in I\}}{\#((\mathcal{S} \setminus \mathbb{N}_0) \cap [N, N + 1))} \longrightarrow |I|$$

as $N \rightarrow \infty$ through values for which the denominator is nonzero.

Proof. By [theorem 9.1](#), the relevant fractional parts are

$$\{\beta\}, \{2\beta\}, \dots, \{K_N\beta\}, \quad K_N = \left\lfloor \frac{N+1}{\beta} \right\rfloor \rightarrow \infty.$$

Since β is irrational, Weyl’s criterion applies. For every nonzero integer h ,

$$\frac{1}{K} \sum_{k=1}^K e^{2\pi i h k \beta} = \frac{e^{2\pi i h \beta} (1 - e^{2\pi i h K \beta})}{K(1 - e^{2\pi i h \beta})} \rightarrow 0.$$

Therefore $\{k\beta\}$ is uniformly distributed modulo one, which gives the claim. $\square$

Status: [new proof in this report]

This is stronger than mere density and gives an exact interpretation of the survey’s missing `FractionalPartDensity` direction. No effective discrepancy rate follows without additional Diophantine information on β ; the geometric-sum proof gives convergence for each Fourier mode but no uniform control over many modes.

9.4 Why equidistribution does not finish the conjecture

For each k , the committed shrinking-target theorem gives approximately

$$\|k\beta\|_{\mathbb{Z}} \gtrsim M^{-k}, \quad M = 2^\beta.$$

Equidistribution guarantees visits to any *fixed* neighborhood of an integer and, with more work, polynomial-scale neighborhoods along subsequences. It does not force $\|k\beta\|_{\mathbb{Z}}$ below an exponentially shrinking radius M^{-k} . Numbers with bounded continued-fraction coefficients, for example, satisfy a much larger lower bound of order $1/k$, and still have equidistributed multiples. The two statements therefore fit together without tension.

10 Sharper finite-difference obstructions

The committed local combinatorial theorem forbids three candidate bases in arithmetic progression under the convenient sufficient condition $h^5 \leq n$. The underlying curvature argument yields a sharper natural exponent and extends to arbitrary progression length.

10.1 An integral representation for finite differences

For a function f and step $h > 0$, write

$$\Delta_h f(t) = f(t+h) - f(t), \quad \Delta_h^r = \underbrace{\Delta_h \cdots \Delta_h}_{r \text{ times}}.$$

If $f \in C^r$, repeated use of the fundamental theorem of calculus gives

$$\Delta_h^r f(n) = \int_{[0,h]^r} f^{(r)}(n + t_1 + \cdots + t_r) dt_1 \cdots dt_r. \quad (6)$$

For $f(t) = t^\vartheta$ and $r \geq 2$,

$$f^{(r)}(t) = c_r^{\text{sgn}} t^{\vartheta-r}, \quad c_r^{\text{sgn}} = \vartheta(\vartheta-1) \cdots (\vartheta-r+1).$$

Because $1 < \vartheta < 2$, this derivative has a fixed nonzero sign on $(0, \infty)$ and its absolute value decreases with t .

Theorem 10.1 (Higher-difference progression obstruction). *Let $r \geq 2$ and $n, h \in \mathbb{N}_{>0}$. Put*

$$c_r = |\vartheta(\vartheta - 1) \cdots (\vartheta - r + 1)|.$$

If all $r + 1$ integers

$$n, n + h, n + 2h, \dots, n + rh$$

are in $\mathcal{C}$, then

$$1 \leq c_r h^r n^{\vartheta-r}.$$

Consequently such a progression is impossible whenever

$$c_r h^r < n^{r-\vartheta}. \quad (7)$$

Proof. If all $n + jh$ are candidates, then every $(n + jh)^\vartheta$ is an integer, and therefore the integer linear combination $\Delta_h^r f(n)$ is an integer. Equation (6) and the fixed sign of $f^{(r)}$ show that this finite difference is nonzero. Hence its absolute value is at least one. Since $t^{\vartheta-r} \leq n^{\vartheta-r}$ for $t \geq n$,

$$1 \leq |\Delta_h^r f(n)| \leq \int_{[0,h]^r} c_r n^{\vartheta-r} dt = c_r h^r n^{\vartheta-r}.$$

Rearranging gives the contrapositive (7). □

Status: [new proof in this report]

The proof is short enough to be an attractive Lean target: it replaces a nested mean-value construction by the iterated-integral identity and reduces the numerical side to monotonicity of a negative real power.

10.2 The sharpened three-term criterion

For $r = 2$, the theorem says that $n, n + h, n + 2h$ cannot all be candidates when

$$\vartheta(\vartheta - 1)h^2 < n^{2-\vartheta}. \quad (8)$$

Equivalently,

$$h < \frac{1}{\sqrt{\vartheta(\vartheta - 1)}} n^{(2-\vartheta)/2} = 1.038547802 \dots n^{0.2075187496 \dots}.$$

The committed hypothesis $h^5 \leq n$ gives $h \leq n^{0.2}$ and is therefore a clean but strictly weaker sufficient condition.

10.3 Longer progressions

The first few thresholds from (7) are shown below. The coefficient $C_r = c_r^{-1/r}$ gives the sufficient exclusion $h < C_r n^{1-\vartheta/r}$.

r	progression length $r + 1$	exponent $1 - \vartheta/r$	C_r
2	3	0.207518750	1.038547802
3	4	0.471679166	1.374849673
4	5	0.603759375	1.164124485
5	6	0.683007500	0.946705202
6	7	0.735839583	0.778537835
7	8	0.773576786	0.652646038
8	9	0.801879687	0.557368999

As r grows, the admissible step exponent approaches one. This looks tantalizing, but it does not immediately contradict the exact candidate monoid. In a dyadic block near size X , the conditional number of candidates is only $O(\log X)$, so the average additive gap in value space is of order $X/\log X$. For each fixed r , this remains larger than $X^{1-\vartheta/r}$. A successful use of long progressions would need r growing with the height together with quantitative control of the rapidly changing coefficient c_r and a combinatorial theorem that actually produces such progressions.

11 Independent finite verification through 2^{20}

11.1 Statement and trust boundary

Proposition 11.1 (Finite exclusion under interval-software trust). *For every integer m with $1 \leq m < 2^{20}$, if $m^\vartheta \in \mathbb{Z}$, then m is a power of two. Equivalently, under the interval computation described below,*

$$2^x, 3^x \in \mathbb{Z}, \quad 2^x < 2^{20} \implies x \in \mathbb{Z}.$$

Thus a nonintegral solution would satisfy $x > 20$.

Status: [\[interval computation\]](#)

The scan used CPython 3.13.5 and `mpmath` 1.3.0 interval arithmetic at 40 decimal digits. It is reproducible and uses outward interval endpoints, but it has not been translated into a Lean certificate and was not replayed by the Lean kernel. It should therefore be viewed as a strong finite computational result, not as an extension of the repository’s trusted theorem `eight_le_of_not_integer_of_two_three_rpow_integer`.

11.2 Certificate inequality

For each non-power of two m , a floating approximation proposes $n = \lfloor m^\vartheta \rfloor$. The proposal is accepted only after interval arithmetic proves both strict inequalities

$$\log m \log 3 - \log n \log 2 > 0, \tag{9}$$

$$\log(n+1) \log 2 - \log m \log 3 > 0. \tag{10}$$

Because $\log 2 > 0$, these are exactly

$$n < m^\vartheta < n + 1.$$

Once both interval lower endpoints are positive, the initial floating approximation has no remaining logical role: even if it had proposed the wrong n , the two certified inequalities could not simultaneously hold.

All 1,048,555 non-powers of two below 2^{20} passed. The scan was split into two equal ranges for reproducibility.

Range	Checked	Failures	Smallest certified log margins
$1 \leq m < 2^{19}$	524,268	0	Lower: $1.4147089634746718 \times 10^{-15}$ at $(m, n) = (189427, 231498127)$; upper: $3.0841010256409534 \times 10^{-15}$ at $(306982, 497586055)$.
$2^{19} \leq m < 2^{20}$	524,287	0	Lower: $9.490232037370244 \times 10^{-16}$ at $(958460, 3023921230)$; upper: $2.3019781003173366 \times 10^{-15}$ at $(556061, 1275861767)$.

At the globally tightest lower-log case, the 40-digit interval had endpoints

$$L = 9.4902320373702458071570103625992802681 \times 10^{-16},$$

$$U = 9.4902320373702458071570232195687423869 \times 10^{-16}.$$

so its width was about 1.3×10^{-54} , roughly thirty-nine decimal orders of magnitude smaller than the positive margin. A separate 100-digit point calculation gives

$$958460^\theta - 3023921230 = 4.1402049868034052578 \times 10^{-6}.$$

The closest lower integer gap among the displayed extremal cases occurs at $m = 189427$ and is approximately $4.7248619698623295 \times 10^{-7}$.

11.3 Manifest hashes

For each checked range, the script hashes the ASCII stream `m:n`; of proposed-and-certified pairs. The resulting SHA-256 digests are

```
02b589461fa8c1721bc251575b98b1e3d58001f54cb7faa2cd6c59edb8c819a5,
fccccf5de8673379d37ff0fc840ab091d9ff7a2352ff64a93091f01c21bf4cb4.
```

These do not constitute proofs by themselves; they make accidental changes to the candidate sequence detectable across reruns.

11.4 Reproducible script

```
1 import math, time, hashlib
2 import mpmath as mp
3
4 START = 1
5 LIMIT = 1 << 19
6 DPS = 40
7
8 mp.iv.dps = DPS
9 ln2 = mp.iv.log(2)
10 ln3 = mp.iv.log(3)
```

```

11 theta_float = math.log(3.0) / math.log(2.0)
12
13 checked = 0
14 failures = []
15 min_lower = (float("inf"), None, None)
16 min_upper = (float("inf"), None, None)
17 manifest = hashlib.sha256()
18 start_time = time.time()
19
20 for m in range(START, LIMIT):
21     if (m & (m - 1)) == 0:      # integral exponent: m is a power of two
22         continue
23
24     # Only proposes a neighboring integer. The interval tests below certify it.
25     n = math.floor(math.exp(theta_float * math.log(m)))
26
27     lhs = mp.iv.log(m) * ln3
28     lower_diff = lhs - mp.iv.log(n) * ln2
29     upper_diff = mp.iv.log(n + 1) * ln2 - lhs
30
31     lower_lb = float(lower_diff.a)
32     upper_lb = float(upper_diff.a)
33     if not (lower_lb > 0.0 and upper_lb > 0.0):
34         failures.append((m, n, str(lower_diff), str(upper_diff)))
35         break
36
37     min_lower = min(min_lower, (lower_lb, m, n))
38     min_upper = min(min_upper, (upper_lb, m, n))
39     manifest.update(f"{m}:{n};".encode())
40     checked += 1
41
42 print("checked_nonpowers=", checked)
43 print("failures=", len(failures))
44 print("min_lower_log_margin=", min_lower)
45 print("min_upper_log_margin=", min_upper)
46 print("sha256_m_n_manifest=", manifest.hexdigest())
47 print("elapsed_seconds=", time.time() - start_time)
48 assert not failures

```

Listing 1: Interval verification for one range. Set START to 1 or 2^{19} and LIMIT to 2^{19} or 2^{20} .

11.5 How to turn this into a kernel certificate

The interval scan is useful reconnaissance, but the repository already demonstrates a stronger trust architecture: generate rational inequalities externally and replay them with exact integer arithmetic in Lean. A scalable 2^{20} formal certificate should separate:

1. a small table identifying, for each m , a certified neighboring integer n and a rational enclosure for ϑ or the relevant logarithms;
2. exact lemmas converting those enclosures into (9)–(10);
3. chunked data modules to keep elaboration and memory bounded;
4. one imported theorem whose trust boundary is only Lean’s kernel and the axioms already used by Mathlib.

A more compact alternative is to use continued-fraction convergents for ϑ and exact integer comparisons m^p versus n^q , sharing one convergent across large ranges. The tradeoff is enormous integers; reflection or verified binary powering is preferable to giant source literals.

12 Attempts at a full proof and their exact failure points

12.1 Attempt 1: collapse rank two to rank one

The committed determinant theorem gives multiplicative rank at most two, and the always-present candidate 2 consumes one direction. It is natural to hope that integrality of m^ϑ forces the remaining odd direction to disappear. The new exact structure theorem shows precisely what remains: a failure would produce a perfectly coherent rank-two monoid

$$\mathcal{C} = \langle 2, M \rangle_{\text{monoid}}.$$

Nothing in the rank argument distinguishes this from the allowed rank-one monoid of powers of two. Excluding the second generator is exactly the original transcendence problem.

12.2 Attempt 2: use density of multiples against endpoint gaps

The orbit $k\beta \pmod{1}$ is not only dense but equidistributed. However, the integrality gap is of exponential scale M^{-k} . Standard equidistribution controls fixed intervals, and even strong classical discrepancy bounds from continued fractions are polynomial in k unless one knows very special partial quotients. There is no contradiction between an irrational rotation and avoiding exponentially small targets. To close this route one would need a theorem specific to the transcendental β satisfying $2^\beta, 3^\beta \in \mathbb{N}$, not generic metric equidistribution.

12.3 Attempt 3: combine exact counts with local progression obstructions

Conditional failure gives $O(\log X)$ candidates in a multiplicative block $[X, 2X)$, with exact constant $1/\beta$. Additive-combinatorial theorems such as Roth’s theorem become powerful for sets of positive density, but $O(\log X)$ points are far below that regime. The new higher-difference obstruction reaches steps $X^{1-\vartheta/r}$ for fixed progression length $r+1$, while the average candidate gap is $X/\log X$. Thus sparse exact structure and local curvature remain mutually consistent.

A genuinely global theorem for S -unit equations among the monomials $2^n M^k$ might still be useful. A three-term progression would solve

$$2^{n_1} M^{k_1} + 2^{n_3} M^{k_3} = 2^{n_2+1} M^{k_2}.$$

General S -unit theory gives finiteness in fixed nondegenerate settings, not existence. The absence of such progressions is therefore not contradictory.

12.4 Attempt 4: exploit the primitive output pair

The pair (M, A) is highly primitive by [theorems 7.3 to 7.5](#). One might hope that these conditions conflict with

$$\frac{\log M}{\log 2} = \frac{\log A}{\log 3}.$$

They do rule out all obvious extractions of common integer shifts and roots, but the remaining equality is still (3). Known lower bounds for linear forms in logarithms do not apply to a product-of-logs determinant with both coefficient rows transcendental.

12.5 Attempt 5: force algebraicity of a forbidden auxiliary exponential

The five exponentials theorem says, in a suitable normalization, that if the four corner exponentials are algebraic, then an additional exponential $e^{\gamma x}$ is transcendental for every nonzero algebraic γ [7, 4]. This is a genuine restriction on a counterexample, but the natural auxiliary quantities needed to reflect prime factorization, such as $p^x = e^{x \log p}$, have coefficient $\log p$, which is transcendental rather than algebraic. The theorem therefore does not directly turn divisors of M or A into a contradiction.

The ordinary six exponentials theorem does imply that if b is a natural base multiplicatively independent of 2 and 3, then b^x must be transcendental: the other five entries in the 3×2 exponential grid are algebraic. The committed `ThreeSmooth` theorem proves the weaker but formalized consequence that b^x cannot be an integer. Again, this does not prohibit b from dividing the integer $M = 2^x$; it only controls the distinct quantity b^x .

12.6 The irreducible remainder

All successful reductions converge on the following statement.

Conjecture 12.1 (Radical four-exponentials target). *For every odd integer $u > 1$,*

$$u^{\log_2 3} \notin \mathbb{Z}[1/3].$$

Equivalently, for every primitive odd $w > 1$ and every $d \geq 1$,

$$\left(3^{\log_2 w}\right)^d \notin \mathbb{Z}[1/3].$$

By [theorem 8.1](#), both formulations are exactly equivalent to [theorem 2.1](#): write an arbitrary odd $u > 1$ as a power of its primitive odd core, or take $u = w^d$ in the reverse direction. The restriction to $\mathbb{Z}[1/3]$ is essential; a rational value with a denominator involving another prime could not be cleared by an integral shift of the exponent. The special case $w = 3, d = 1$ asks whether $3^{\log_2 3}$ lies outside $\mathbb{Z}[1/3]$; even the stronger assertion that it is irrational is already outside present Gelfond–Schneider and Baker methods. Thus a proof that treats this special case by a genuinely new mechanism would be a meaningful breakthrough even before the full uniform theorem.

13 Prioritized future research directions

The following priorities are ordered by a mixture of mathematical leverage and formalization feasibility.

13.1 Priority A: formalize the exact primitive-generator theorem

This should be the immediate next repository milestone. It converts three article-level threads—odd core, fractional-part density, and exponent semigroup—into one canonical structure theorem and gives many corollaries almost for free.

A plausible module split is:

OddCore.lean Construct the primitive common odd core from the committed cross-valuation theorem `odd_factorization_cross_mul`; prove uniqueness and the representation $m = 2^i w^j$.

RationalPowerIndex.lean For $E > 0$, define the subgroup of integer exponents with rational powers; identify it with $d\mathbb{Z}$; prove the denominator-support lemma for E^d .

PrimitiveGenerator.lean Construct β , M , and A and prove $\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta$ and $\mathcal{C} = \langle 2, M \rangle$.

ExactCounting.lean Prove the unit-block and cumulative formulas.

FractionalPartEquidistribution.lean Prove Weyl’s criterion for one irrational rotation via the finite geometric sum, then derive blockwise equidistribution.

The hardest engineering points are likely to be: a gcd over the finite support of a `Finsupp`; a Bézout certificate showing a rational scalar is integral; converting the subgroup of $\mathbb{Z}$ into a least positive generator; and bookkeeping between natural, integer, rational, and real powers. None requires new transcendence theory.

13.2 Priority B: formalize the sharper finite differences

A reusable theorem for iterated forward differences and iterated integrals would benefit more than this project. Once (6) is in place, the candidate obstruction is a short specialization. The $r = 2$ theorem should replace $h^5 \leq n$ by the sharp natural condition (8); the arbitrary- r theorem can follow afterward.

Potential Lean-facing statement:

```
theorem no_candidate_arithmetic_progression_of_finiteDifference_lt_one
  (r n h : Nat) (hr : 2 <= r) (hn : 0 < n) (hh : 0 < h)
  (hsmall : fallingAbs theta r * h^r < n^(r - theta)) :
  not (forall j : Fin (r + 1), TwoBaseNaturalCandidate (n + j * h))

theorem no_three_term_candidate_AP_sharp
  (hsmall : theta * (theta - 1) * h^2 < n^(2 - theta)) :
  not (Candidate n and Candidate (n+h) and Candidate (n+2*h))
```

Listing 2: Schematic theorem signatures; not compiled Lean.

The exact syntax will differ because real powers and casts need explicit handling.

13.3 Priority C: kernel-replay the 2^{20} finite region

The interval computation has already identified a clean target and the worst cases. The next step is not a larger untrusted scan; it is a compact proof object. A generator can choose continued-fraction enclosures adaptively, emit chunked rational certificates, and have Lean replay only exact inequalities. The target theorem would be

```
theorem integer_of_two_three_rpow_integer_of_two_rpow_lt_1048576
  {x : Real}
  (h2 : (2 : Real)^x in Set.range ((up) : Int -> Real))
  (h3 : (3 : Real)^x in Set.range ((up) : Int -> Real))
  (hlt : (2 : Real)^x < 1048576) :
  x in Set.range ((up) : Int -> Real)
```

After this, every descent theorem’s constant improves from 8 to 20 by bookkeeping.

13.4 Priority D: isolate the radical target in a reusable API

The equivalence in [theorem 8.1](#) should be formalized before attacking it. Useful interfaces include:

- failure of the conjecture $\iff$ an odd $u > 1$ with $u^v \in \mathbb{Z}[1/3]$;
- failure $\iff$ existence of a primitive odd w , $d \geq 1$, and $q \in \mathbb{Z}[1/3]_{>0}$ with $(3^{\log_2 w})^d = q$;
- minimal $d \implies$ irreducibility of $X^d - q$;
- a symmetric localization at 2 obtained by swapping the bases.

This creates a small, sharply stated endpoint for importing future transcendence results.

13.5 Priority E: five and sharp six exponentials

Formalizing the five exponentials theorem or Roy’s sharp six exponentials theorem would add new restrictions such as transcendence of $e^{\gamma x}$ for nonzero algebraic γ . It is not by itself a proof of the conjecture, but it would enlarge the formal transcendence library and may interact with specially constructed auxiliary functions. The effort is much larger than Priorities A–D and should be undertaken with a specific downstream lemma in mind, not merely for inventory completeness.

13.6 Priority F: auxiliary-function methods tailored to integrality

A route beyond four exponentials must exploit more than algebraicity. Here all four corner values are positive integers, all their natural powers remain integers, and the exact monoid theorem provides a canonical primitive pair. An auxiliary polynomial or determinant could be designed to vanish to high order on

$$\{(2^n M^k, 3^n A^k) : 0 \leq n < N, 0 \leq k < K\}$$

while incorporating divisibility or congruence information unavailable in the general four-exponentials setting. The danger is that this grid is itself a perfectly regular rank-two multiplicative group; a successful construction needs a new arithmetic lower bound stronger than “nonzero algebraic integer has absolute value at least one.”

Promising variants include:

- use several non-Archimedean valuations simultaneously, not only the real absolute value;
- exploit the primitive exclusions in [theorem 7.5](#) to prevent determinant factors from sharing large perfect powers;
- work with a resultant or norm in the radical field $\mathbb{Q}(E)$ from [theorem 8.2](#), combining Archimedean smallness with denominator support only at 3;
- search for a product formula contradiction across all places of $\mathbb{Q}(E)$.

The last idea is the most structurally aligned with the new radical formulation: E has a binomial minimal polynomial and tightly controlled denominator, so all conjugates and local heights are explicit.

13.7 Priority G: primitive-pair computation rather than raw candidate scanning

The exact theorem suggests a more focused experimental search. A counterexample has a primitive pair (M, A) satisfying

$$M^\vartheta = A, \quad \min(v_2(M), v_3(A)) = 0,$$

with no common perfect-power degree and no nontrivial matched affine decomposition. Finite searches can reject candidate M as soon as one of these conditions fails, and can record the continued-fraction or logarithmic near miss in a database keyed by the odd core. Such a search will not prove the infinite statement, but it can reveal whether the closest approaches cluster around particular smoothness patterns, denominators, or rational-power indices d .

13.8 Priority H: quantitative irrationality and discrepancy

Effective irrationality measures for ϑ improve finite certification of m^ϑ , but they do not directly control the primitive generator $\beta = \log_2 M$, whose integer M is unknown. Conversely, generic discrepancy for $k\beta$ does not reach the exponentially shrinking target. This direction becomes decisive only if one can derive a special effective measure for logarithms of the hypothetical output pair from the relation $\log M \log 3 = \log A \log 2$. Any claimed polynomial measure should be checked carefully against the fact that this relation lies beyond ordinary linear forms in logarithms.

14 Lean formalization blueprint

This section spells out the proof dependencies in a form suitable for implementation planning. The displayed Lean is schematic; names and coercions must be adapted to the repository.

14.1 Definitions

```
def TwoBaseSolution (x : Real) : Prop :=
  (2 : Real)^x in Set.range ((up) : Int -> Real) and
  (3 : Real)^x in Set.range ((up) : Int -> Real)
```

```

def TwoBaseNaturalCandidate (m : Nat) : Prop :=
  0 < m and exists n : Nat, (n : Real) = (m : Real)^logThreeDivLogTwo

def IsPrimitiveOddCore (w : Nat) : Prop :=
  Odd w and 1 < w and
  Nat.gcd (w.factorization.support.image w.factorization) = 1

```

The actual code already defines the candidate predicate. A robust primitive-core definition should avoid taking a gcd of an empty finset by bundling $w > 1$ and should probably work directly with the finite support excluding 2.

14.2 Odd-core construction

The core proof can be decomposed into small lemmas:

1. choose an anchor candidate with an odd prime factor;
2. define g as the gcd of its positive odd valuations;
3. define w by dividing each odd valuation by g and reconstructing a natural number from the resulting `Finsupp`;
4. use `odd_factorization_cross_mul` to show every candidate odd valuation vector is a rational scalar multiple of the primitive vector;
5. use a finite Bézout combination of the primitive coordinates to prove the scalar is a natural number;
6. reconstruct $m = 2^i w^j$ by equality of all prime factorizations;
7. prove uniqueness.

Mathlib may not expose exactly the desired “gcd of a finset has a Bézout combination” lemma. A practical route is to induct over the finite support while carrying coefficients, or to use `Ideal.span` in $\mathbb{Z}$ and identify the ideal generated by the coordinates with the unit ideal.

14.3 Rational-power subgroup

Let $E = 3^{\log_2 w} > 0$. Define

$$J = \{j \in \mathbb{Z} : E^j \in \mathbb{Q}\}.$$

In Lean, it may be simpler to define an `AddSubgroup Int` using integer powers in $\mathbb{R}^\times$ than to mix `Real.rpow` with negative integer casts. Positivity makes ordinary integer powers available. Then use the classification of additive subgroups of $\mathbb{Z}$, or prove directly that a nonzero subgroup has a least positive element dividing every member.

The denominator lemma can be stated purely for rationals:

```

theorem rat_den_prime_support_of_pow_mul_three_pow_integer
  {q : Rat} (hq : 0 < q) {i k : Nat} (hk : 0 < k)
  (h : exists z : Nat, (3 : Rat)^i * q^k = z) :
  exists a A : Nat, q = A / 3^a and (a = 0 or not 3 dvd A)

```

A valuation proof over `Rat.num` and `Rat.den` is likely shorter than developing localization machinery.

14.4 Exact representation theorem

Once $E^d = A/3^a$ is available, the central calculation is elementary:

$$3^i E^{dk} = 3^{i-ak} A^k \in \mathbb{N} \iff i \geq ak.$$

The reverse implication is direct. The forward implication requires reducedness of the rational and a divisibility argument. The rest is normalization and irrational uniqueness.

A useful final signature is:

```
theorem exists_unique_primitive_generator
  (hcounter : exists x : Real, TwoBaseSolution x and not IsInteger x) :
  exists beta M A,
    not IsInteger beta and
    (M : Real) = (2 : Real)^beta and
    (A : Real) = (3 : Real)^beta and
    (forall x, TwoBaseSolution x <->
      exists! nk : Nat x Nat, x = nk.1 + nk.2 * beta) and
    (forall m, TwoBaseNaturalCandidate m <->
      exists! nk : Nat x Nat, m = 2^nk.1 * M^nk.2)
```

It may be cleaner to split existence, classification, and uniqueness into separate theorems rather than force dependent uniqueness into one statement.

14.5 Equidistribution without a large library dependency

For the specific sequence $k\beta$, the Weyl sums are finite geometric series. One can prove:

```
theorem geometric_weyl_sum_tendsto_zero
  (hbeta : Irrational beta) (h : Int) (hh : h != 0) :
  Tendsto (fun K => (1 / K) *
    sum k in Finset.Icc 1 K, Complex.exp (2 * pi * I * h * k * beta))
  atTop (nhds 0)
```

The remaining infrastructure is the Weyl criterion or a direct approximation of interval indicators by trigonometric polynomials. If Mathlib lacks the full criterion, formalizing it is a meaningful reusable contribution. Density alone is much easier, but the exact block structure deserves the stronger result.

15 Assessment of the survey’s proposed directions

The existing survey proposes seven directions. The new analysis changes their priority as follows.

15.1 Equidistribution

This direction is mathematically complete conditional on failure: [theorem 9.2](#) proves Weyl equidistribution by a one-line geometric-sum estimate. What remains is formalization and, separately, effective discrepancy. The latter still does not confront the exponential shrinking target without special Diophantine input.

15.2 Five exponentials

Still worthwhile as library development and as a source of auxiliary transcendence statements, but not a direct route to p^x because $\log p$ is not algebraic. A project should identify a specific algebraic coefficient γ whose exponential $e^{\gamma x}$ can also be forced algebraic from the primitive pair; otherwise the theorem only adds a restriction that is easy to state and hard to use.

15.3 Effective irrationality measures for ϑ

Highly useful for scaling finite candidate checks and replacing enormous per-value certificates. Less directly useful for the exact primitive generator, which involves $\log_2 M$ for an unknown integer M . The finite-check application is concrete and should be pursued.

15.4 Larger verified regions

The independent scan shows that 2^{20} is computationally routine. The bottleneck is no longer discovering the finite truth but expressing it in a compact kernel-replayable form. Certificate architecture should take priority over simply extending an untrusted endpoint.

15.5 Sharper local combinatorics

Partly completed here: [theorem 10.1](#) supplies the sharper three-term exponent and all higher orders. The remaining open issue is global combinatorial leverage on an $O(\log X)$ -sparse set.

15.6 Exponent semigroup structure

Substantially resolved by [theorem 7.2](#). There are no arbitrary Frobenius-type gaps: all solutions are $n + k\beta$. Future work should use this exact form rather than study a general submonoid $B \subseteq \mathbb{N}_0^2$.

15.7 Colossally abundant numbers

Historically important and suitable for formalization, but logically downstream. It explains why Alaoglu and Erdős cared about the transcendence statement; it does not currently feed back into a proof. It should be pursued as an application after the core structure modules are stabilized.

16 Conclusions

The requested commits contain a serious and unusually broad formal attack on the Alaoglu–Erdős conjecture. The strongest verified themes are complementary: interpolation determinants settle three independent bases; Gelfond–Schneider excludes algebraic irrational exponents; finite certificates create a zero-free region; prime factorization bounds multiplicative rank; finite differences force local additive sparsity; and affine descent constrains perfect-power structure.

The full conjecture does not fall to a simple combination of those results. The reason can now be stated very sharply. If a counterexample exists, it does not generate a messy exceptional set that density or semigroup theory is likely to squeeze accidentally. It generates the perfectly rigid free monoid

$$\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta, \quad \mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\}.$$

Its block counts, fractional parts, and multiplicative relations are all explicit and internally consistent with the committed gap and density bounds. The remaining assertion is that the second generator M cannot exist, equivalently that no odd $u > 1$ has $u^{\log_2 3} \in \mathbb{Z}[1/3]$.

That reformulation is progress rather than pessimism. It removes several artificial degrees of freedom, supplies a canonical primitive object, turns article-level density claims into exact theorems, and gives a focused interface for future transcendence methods. The most productive near-term program is therefore:

1. formalize the primitive generator and exact counting;
2. kernel-replay the 2^{20} finite region;
3. formalize the sharp finite-difference family;
4. investigate product-formula and auxiliary-determinant arguments in the explicit radical field $\mathbb{Q}(3^{\log_2 w})$ under the primitive denominator constraints.

A complete proof still requires a new idea at the four-exponentials boundary, but the target is now considerably narrower and the surrounding formal infrastructure is strong enough to test such an idea precisely.

A Status matrix for major statements

Statement	Status in this report	Location
$2^x, 3^x, 5^x \in \mathbb{Z} \Rightarrow x \in \mathbb{Z}$	[committed Lean]	IntegerExponent.lean
Nonintegral solution is irrational/transcendental	[committed Lean]	TwoBaseIntegerExponent.lean, Transcendence.lean
$2^x < 256 \Rightarrow x \in \mathbb{Z}$	[committed Lean]	FiniteCheck256.lean
$2^x < 4096 \Rightarrow x \in \mathbb{Z}$	[article-only at the audited commit]	Survey marks final Lean acceptance pending; named module absent
Candidate prime-exponent rank at most two	[committed Lean]	MultiplicativeRank.lean
Common primitive odd core	[new proof in this report]	Theorem 6.2; based on committed cross-valuation lemma
Exact solution monoid $\mathbb{N}_0 + \mathbb{N}_0\beta$	[new proof in this report]	Theorem 7.2
Exact dyadic-block counts	[new proof in this report]	Theorem 9.1
Blockwise equidistribution	[new proof in this report]	Theorem 9.2
3-smooth classification of additional integral bases	[committed Lean]	ThreeSmooth.lean
Affine descent and $r + 8k \leq x$	[committed Lean]	ArithmeticRigidity.lean

Statement	Status in this report	Location
Exact affine primitivity of least generator	[new proof in this report]	Theorem 7.5
Three-term AP obstruction under $h^5 \leq n$	[committed Lean]	ZeroDensity.lean
Sharp and higher-order AP obstructions	[new proof in this report]	Theorem 10.1
No nonintegral candidate with $2^x < 2^{20}$	[interval computation]	Theorem 11.1; not Lean checked
Full Alaoglu–Erdős conjecture	Open in this report	Reduced to theorem 12.1

B Detailed interval outputs

The two 40-digit runs produced:

```
LIMIT=524288
DPS=40
checked_nonpowers=524268
failures=0
min_lower_log_margin=(1.4147089634746718e-15, 189427, 231498127)
min_upper_log_margin=(3.0841010256409534e-15, 306982, 497586055)
sha256_m_n_manifest=02b589461fa8c1721bc251575b98b1e3d58001f54cb7faa2cd6c59edb8c819a5
elapsed_seconds=18.505
```

Listing 3: Lower half output.

```
LIMIT=1048576
DPS=40
checked_nonpowers=524287
failures=0
min_lower_log_margin=(9.490232037370244e-16, 958460, 3023921230)
min_upper_log_margin=(2.3019781003173366e-15, 556061, 1275861767)
sha256_m_n_manifest=fcccf5de8673379d37ff0fc840ab091d9ff7a2352ff64a93091f01c21bf4cb4
elapsed_seconds=18.904
```

Listing 4: Upper half output.

Elapsed times are machine-dependent and are included only as a sanity check. The counts, zero-failure result, extremal tuples, and hashes are the reproducibility data.

C A symmetric formulation

Everything above has a base-swapped version. Put $\eta = \log_3 2 = 1/\vartheta$. Failure is also equivalent to the existence of an integer $v > 1$ with $3 \nmid v$ such that

$$v^\eta \in \mathbb{Z}[1/2].$$

For the primitive pair (M, A) , factoring A into a power of 3 times a 3-coprime part produces a symmetric primitive core and rational-power index. The two normalizations give the same least exponent β . The compatibility condition between them is exactly

$$\log M \log 3 = \log A \log 2.$$

This symmetry is useful in formalization because some divisibility arguments may be easier on one side, and it confirms that [theorem 7.3](#) is not an artifact of privileging the base 2.

D Bibliographic and repository references

References

- [1] L. Alaoglu and P. Erdős, *On highly composite and similar numbers*, Transactions of the American Mathematical Society **56** (1944), 448–469. DOI: [10.1090/S0002-9947-1944-0011087-2](https://doi.org/10.1090/S0002-9947-1944-0011087-2).
- [2] A. Baker, *Transcendental Number Theory*, Cambridge University Press, 1975.
- [3] L. Kuipers and H. Niederreiter, *Uniform Distribution of Sequences*, Wiley, 1974; Dover reprint, 2006.
- [4] D. Roy, *Matrices whose coefficients are linear forms in logarithms*, Journal of Number Theory **41**(1) (1992), 22–47. DOI: [10.1016/0022-314X\(92\)90081-Y](https://doi.org/10.1016/0022-314X(92)90081-Y).
- [5] Th. Schneider, *Einführung in die transzendenten Zahlen*, Springer, 1957.
- [6] M. Waldschmidt, *Diophantine Approximation on Linear Algebraic Groups: Transcendence Properties of the Exponential Function in Several Variables*, Grundlehren der mathematischen Wissenschaften 326, Springer, 2000. DOI: [10.1007/978-3-662-11569-5](https://doi.org/10.1007/978-3-662-11569-5).
- [7] M. Waldschmidt, *Variations on the Six Exponentials Theorem*, in *Algebra and Number Theory*, pp. 338–355. DOI: [10.1007/978-93-86279-23-1_23](https://doi.org/10.1007/978-93-86279-23-1_23).
- [8] M. Waldschmidt, *The Four Exponentials Problem and Schanuel’s Conjecture*, in *Mathematics Going Forward*, Lecture Notes in Mathematics, 2023, pp. 579–592. DOI: [10.1007/978-3-031-12244-6_39](https://doi.org/10.1007/978-3-031-12244-6_39).
- [9] TheoremDB contributors, *Four exponentials conjecture*, problem P4, reviewed July 31, 2026. <https://theoremdb.org/statements/P4/>.
- [10] V. Reshetnikov, *ProveIt repository*, <https://github.com/VladimirReshetnikov/ProveIt>.
- [11] V. Reshetnikov, Commit `f492c18f1f2f630fd6c70e3d6ecce4a0229f7bc2`, *Formalize integer real exponent theorem*, August 20, 2026.
- [12] V. Reshetnikov, Commit `a37d97f9d43297c89e439aeb68d55406fe8c1eb1`, *Formalize partial results for two-base integer exponents*, August 20, 2026.
- [13] V. Reshetnikov, Commit `4a9b03d12daa7a51992d0106c0e6ec9eb8fef30b`, *Formalize strong partial results for two-base integer exponents*, August 20, 2026.
- [14] V. Reshetnikov, Commit `423ac24f0fec4a44231f19864d6d9c3099cafa3d`, *Survey partial results on the two-base integer-exponent conjecture*, August 20, 2026.
- [15] F. Johansson et al., *mpmath: a Python library for arbitrary-precision floating-point arithmetic*, version 1.3.0 used for the interval computation in this report. <https://mpmath.org/>.